

### Annotated Bibliography:

This annotated bibliography uses sources that focus on collecting and defining water polo data, modeling the probability of shot success, and making data useful for coaches and players. These studies have methodologies and models that can be used as frameworks for my project. My project differs from this collection of sources in that I will be combining manual data sourcing, variable coding, modeling, and a visual dashboard into one project.

The main source that closely aligns with my study is Gullikson et al. (2023). This study runs a statistical analysis of collected manual data from NCAA men's and women's water polo game film. The methodology for data collection is very similar to my own, which will help me determine how to define and code variables such as shot type, shooter positions, and power play situations that can alter a power play shooting scenario.

Another source, Canossa et al. (2020), provides another direct example of a water polo shooting statistical analysis. This study will similarly help me categorize variables like shot location and power play and man-up situations, as well as help me test the reliability of my manual data and coding.

Mead et al. (2023) is not directly related to water polo, however, the fundamentals of the study can be applied to my own statistical water polo analysis. The study is a soccer analysis that uses an expected-goals model, providing a framework for estimating the probability that a shot results in a goal based on the variables and factors of that shooting scenario.

Lastly, Goudsmit et al. (2022) provides a dashboard framework that can help me communicate my findings with coaches. The dashboard will provide data that is useful for coaches and players to learn from.

Gullikson, J., Gale, L. R., Mayberry, J. K., Kim, J., & Killick, L. (2023). "Production functions of NCAA men and women water polo matches." *Journal of Sports Analytics*, 9(2), 141–155. doi:10.3233/JSA-220600.

This source is most similar to my project. The researchers observed data from game film and manually collected and coded the data. They used game film from NCAA women's and men's water polo games. The researchers defined and coded variables for shot location, offensive situations, shot clock time, the opponent's defensive situations, and if the shot was a

goal. This study analyzes the relationship between these different variables and winning games. I plan to use this source as a reference for data collecting and data definition and coding.

Canossa, S., Abrales, J. A., Estriga, L., Fernandes, R. J., & Garganta, J. (2020). "Water Polo Shooting Performance: Differences Between World Championship Winning, Drawing and Losing Teams." *Journal of Human Kinetics*, 72, 203–214. doi:10.2478/hukin-2019-0107.

This study analyzes high-level water polo games and identifies the technical variables that are related to shots that turn into goals. The analysis measures shot location, goal location, power play situations, defensive pressure, and if the shot was successful. The researchers of this analysis tested how reliable the observable data was from game film, as there is observational bias when spectators are documenting statistics. Similarly to the last source, this analysis will give me reference for what variables to use and how to use them. I will be able to define meaningful categories. Additionally, this gives me a reference for a model that tests if subjective statistics are being recorded reliably and consistently.

Mead, J., O'Hare, A., & McMenemy, P. (2023). "Expected goals in football: Improving model performance and demonstrating value." *PLOS ONE*, 18(4), e0282295. doi:10.1371/journal.pone.0282295.

This source is not directly related to water polo, as it is a soccer analysis. However, this study offers me a method to model expected goals. This model estimates the probability, measured zero to one, that a shot will be a goal using information about the pre-shot situation. The researchers look at variables that describe both the shot and context of the game. Using these variables, they compare statistical and machine learning models, including logistic regression models. I plan to adapt this expected goals model for Pomona-Pitzer Water Polo power plays. I will estimate the probability that a power play shot will turn into a goal based on contextual variables of the power play situation.

Goudsmit, J., Otter, R. T. A., Stoter, I., van Holland, B., van der Zwaard, S., de Jong, J., & Vos, S. (2022). "Co-Operative Design of a Coach Dashboard for Training Monitoring and Feedback." *Sensors*, 22(23), 9073. doi:10.3390/s22239073.

This analysis translates sports data into a visual dashboard, which is immensely valuable for communicating findings to non-technical audiences like coaches and players. This source is interesting as it involves coaches in the design process of the dashboard, allowing coach input on what is helpful for researchers to include and exclude in the dashboard. This aligns with the goal of my project which is to ultimately turn statistical findings into clear and understandable learnings for the Pomona-Pitzer Water Polo team and program.

#### Data Source:

The primary dataset for this project is not from an actual published dataset. I will build a dataset by coding Pomona-Pitzer Water Polo film. Each power play shot will be one observation, with variables that describe factors like shooter position, shot clock time during the power play, the shot type, shot location, shot direction, shot preparation (i.e. faked, quick shot, etc.), opponent, and shot outcome. The game film that I will use is owned by the Pomona-Pitzer Water Polo coaches and will be accessed for this project. I may also use external information and expertise in measuring opponent strength, rather than relying solely on the division (I, II, III) of the opponent. Official SCIAC rankings and NCAA rankings can provide additional information on the opponent's ranking as well. These variables can be control variables in the shot-success model.